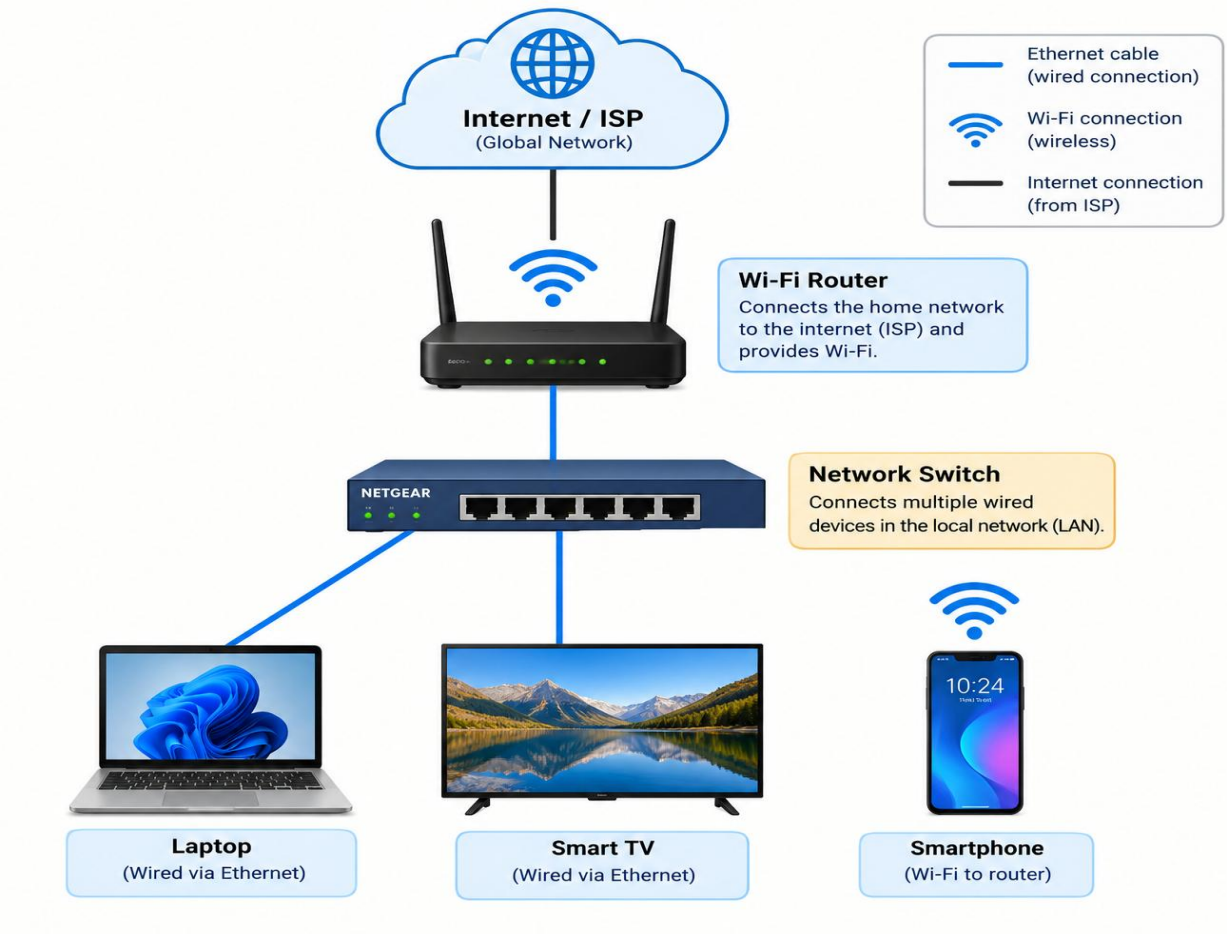
Network Devices

Task 1 : Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.



- A home network connects multiple devices so they can communicate and access the internet.
- The Wi-Fi router connects the home network to the Internet Service Provider (ISP), while the network switch provides additional Ethernet ports for wired devices such as a laptop and smart TV.
- The smartphone connects wirelessly to the router using Wi-Fi.
- All three devices can access internet services through the router.

Task 2 : Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

Network devices are hardware components that connect computers, smartphones, and other devices to a network. Different devices perform different functions to enable communication and data transfer.

Comparison Table

Network Device	Main Function	Advantages	Limitations	Real-World Use Case
Router	Connects different networks and forwards data packets between them.	Connects a LAN to the internet; supports routing and network management.	Incorrect configuration or router failure can interrupt internet access.	Used at home to connect phones and laptops to the internet for browsing, YouTube, and online gaming.
Switch	Connects devices within a LAN and forwards Ethernet frames to the correct port using MAC addresses.	Efficient communication; reduces unnecessary traffic compared with a hub.	Requires additional ports and cabling for wired devices; switch failure can affect connected devices.	Used in an office or computer lab to connect computers for file sharing and LAN gaming.
Hub	Connects multiple devices and broadcasts incoming data to all other ports.	Simple to use and inexpensive in basic setups.	Creates unnecessary traffic; shares bandwidth; offers poor performance and security.	Mainly found in older Ethernet networks or used for basic networking demonstrations.

Access Point (AP)	Provides Wi-Fi connectivity and allows wireless devices to join a wired LAN.	Extends wireless coverage; supports multiple Wi-Fi devices.	Performance depends on signal strength, interference, and available bandwidth.	Used at home, college, or a café to connect smartphones and laptops to Wi-Fi for streaming YouTube or using online services.

Conclusion

- Routers connect different networks, switches efficiently connect wired devices within a LAN, hubs broadcast data to all connected ports, and access points provide wireless connectivity.
- These devices work together to make modern home, office, and college networks possible.

Task 3 : Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

Scenario

When I open the Zomato app on my smartphone and order food, several network devices help transfer data between my phone and Zomato's servers.

Data Transmission Process

1. **Smartphone:** I open Zomato, select a restaurant, choose food, and place my order. The app creates a request containing the required order information.
2. **Wi-Fi Access Point:** If I use Wi-Fi, the access point wirelessly connects my smartphone to the local network.
3. **Router:** The router forwards my data toward the internet and connects my home network to the Internet Service Provider (ISP).
4. **Switch:** Switches in the local network or ISP infrastructure forward data between connected devices and network equipment. My home network may use a switch if additional wired connections are needed.
5. **ISP and Internet:** The ISP carries my request through the internet toward Zomato's servers.
6. **Zomato Server:** The server processes my request and communicates with the relevant services to handle the order. It sends a response confirming the order status.
7. **Response to Smartphone:** The response travels back through the internet and my network connection. My phone displays the confirmation and order details.

Task 4 : Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.

Hint: Think about devices like switches, powerline adapters, or WiFi extenders.

Scenario

A Wi-Fi router is available at home, but my gaming PC is located far away from the router and has only an Ethernet port. I can use the following two network devices to connect my PC to the internet.

1. Powerline Ethernet Adapter

A powerline adapter uses the existing electrical wiring in a building to carry network data between two adapters.

How it works:

- Connect the first adapter to the router using an Ethernet cable and plug it into a wall socket.
- Plug the second adapter into a wall socket near the gaming PC.
- Connect the second adapter to the PC using an Ethernet cable.

Advantages: It can provide a wired connection without running a long Ethernet cable through the house.

Limitations: Speed and stability depend on the quality of the electrical wiring and the electrical circuits.

2. Wi-Fi Extender with an Ethernet Port

A Wi-Fi extender receives the existing router's Wi-Fi signal and retransmits it to improve coverage. Some models also provide an Ethernet port for wired devices.

How it works:

- Place the extender at a location where it receives a good signal from the router.
- Connect the extender to the router's Wi-Fi network.
- Connect the gaming PC to the extender using an Ethernet cable.

Advantages: It avoids running a long cable and can extend wireless coverage to a distant room.

Limitations: Performance depends on signal strength, interference, and the extender's capabilities. Some extenders may increase latency.

Comparison

Feature	**Powerline Adapter**	**Wi-Fi Extender with Ethernet**
Connection method	Existing electrical wiring	Wireless connection to router
PC connection	Ethernet cable	Ethernet cable
Main benefit	Avoids a long Ethernet cable	Extends Wi-Fi coverage
Main limitation	Depends on electrical wiring	Depends on Wi-Fi signal quality
Gaming suitability	Good if the powerline link is stable	Good if the extender receives a strong signal